

Assessing predictions

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Last time: assessing binary predictions

Thresholding: $\hat{Y}_i = \begin{cases} 1 & \hat{p}_i \geq c \\ 0 & \hat{p}_i < c \end{cases}$

Confusion matrix:

	$Y = 0$	$Y = 1$
$\hat{Y} = 0$	True Negatives (TN)	False Negatives (FN)
$\hat{Y} = 1$	False Positives (FP)	True Positives (TP)

Last time

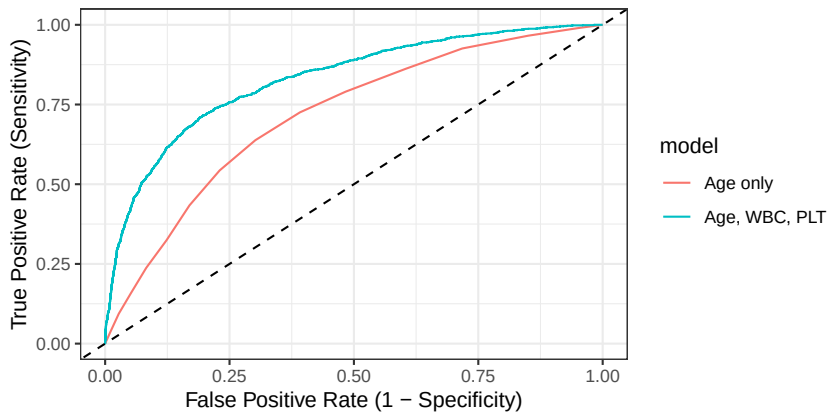
Predicted Status	Actual Status	
	negative	positive
negative	3069	430
positive	954	1267

Predicted Status	Actual Status	
	negative	positive
negative	2327	239
positive	1696	1458

Predicted Status	Actual Status	
	negative	positive
negative	3634	765
positive	389	932

Changing thresholds

ROC Curves



Another example: loan defaults

Data on 948 real estate loans from CA, which were guaranteed by the US Small Business Administration (SBA). Variables include:

- ▶ **Default**: whether the business defaulted on the loan (1 = yes, 0 = no)
- ▶ **UrbanRural**: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- ▶ **NewExist**: 1 if business already existed, 2 if business is new, 0 if unknown
- ▶ **Term**: Length of the loan term (months)
- ▶ **NoEmp**: Number of employees of the business before receiving the loan
- ▶ **GrAppv**: Gross amount of loan approved by the bank, in dollars
- ▶ **SBA_Appv**: Amount of the loan guaranteed by the SBA, in dollars
- ▶ **DisbursementGross**: The amount of money disbursed (loaned), in dollars

Another example: loan default

Fit a model on the 948 loans:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	6.225	1.055	5.899	0.000
log(GrAppv)	-0.154	0.101	-1.521	0.128
log(NoEmp + 1)	0.059	0.126	0.466	0.641
Term	-0.074	0.005	-15.210	0.000

Default? (predicted)	Default? (actual)	
	no	yes
no	438	82
yes	92	336

Predicting on new data?

Predictions on the original 948 loans used to fit the model:

Default? (predicted)	Default? (actual)	
	no	yes
no	438	82
yes	92	336

Acc. = 0.816, Sens. = 0.804, Spec. = 0.826, Deviance = 836.06

Predictions on a **new** set of 407 loans (not used to fit model):

Default? (predicted)	Default? (actual)	
	no	yes
no	165	53
yes	44	145

Acc. = 0.762, Sens. = 0.732, Spec. = 0.790, Deviance = 405.78

Predicting on new data

In general: A model predicts **better** on the data used to fit the model than it does on **new** data.

Why?

Class activity

Work on the class activity (course website)

Terminology and notation

Loss functions

Risk (error)